Hints: Integrals – 2. Symmetry, the king's rule and periodicity

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Substitute $x \mapsto -x$: the integral becomes $\int_{-a}^{a} \frac{f(x)}{1+e^{-x}}dx = \int_{-a}^{a} \frac{e^x f(x)}{1+e^x}dx$. Add it to the original: you get $\int_{-a}^{a} f = 2\int_0^a f$.

Hint 2. $\int_{-1}^{1} \frac{x^2}{1+e^x}dx = \int_0^1 x^2 dx = \frac{1}{3}$ and $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x}dx = \int_0^{\pi/2} \cos x\, dx = 1$.

Problem 2

Hint 1. Pair the terms: $\frac{1}{1+\tan x} + \frac{1}{1+\cot x}$, $\frac{1}{1+\cos x} + \frac{1}{1+\sec x}$, $\frac{1}{1+\sin x} + \frac{1}{1+\csc x}$. Each pair has the form $\frac{1}{1+u} + \frac{1}{1+1/u}$.

Hint 2. $\frac{1}{1+u} + \frac{u}{u+1} = 1$, so the integrand equals 3 wherever it is defined, and the integral is $3x + C$.

Problem 3

Hint 1. The shift $x \mapsto x + \pi$ sends $\sin x \mapsto -\sin x$. So $\int_0^{2\pi} \lfloor 2023 \sin x \rfloor dx = \int_0^{\pi} (\lfloor y \rfloor + \lfloor -y \rfloor)\, dx$ with $y = 2023 \sin x$.

Hint 2. $\lfloor y \rfloor + \lfloor -y \rfloor = -1$ unless y is an integer, which happens only at finitely many x. So the answer is $-\pi$.

Problem 4

Hint 1. Compute the first few nested expressions separately for $x \geqslant 0$ and $x < 0$. For the first one, $g_1 = |x|$, $g_{k+1} = |x + g_k|$: for $x \geqslant 0$, $g_k = kx$; for $x < 0$ the values alternate $-x, 0, -x, 0, \ldots$.

Hint 2. With 2026 (even) occurrences, $g_{2026} = 0$ for $x < 0$ and $2026x$ for $x \geqslant 0$, so the first integral is $\int_0^1 2026x\, dx = 1013$.

Hint 3. For the second, the function is even; for $x \geqslant 0$ let $h_0 = x$, $h_k = |h_{k-1} - 1|$. By induction, on $[0, n]$ the function h_n is a zigzag between 0 and 1 with slopes ± 1 (a triangle wave), so its integral over $[0, n]$ is $\frac{n}{2}$. Hence the answer is $2 \cdot \frac{2023}{2} = 2023$.

Problem 5

Hint 1. In the first integral $x^{2024}dx$ is, up to the factor $\frac{1}{2025}$, exactly $d(x^{2025})$; so $u = x^{2025}$ (or $u = 1 - x^{2025}$, which maps $[0, 1]$ to itself reversed) reduces it to a single power.

Hint 2. First: $\frac{1}{2025}\int_0^1 u^{2025}du = \frac{1}{2025 \cdot 2026}$. Second: $u = 1 - x$ gives $\int (u^{2021} - u^{2020})\, du$ up to sign, so the answer is $\frac{(1-x)^{2022}}{2022} - \frac{(1-x)^{2021}}{2021} + C$.

Problem 6

Hint 1. $\cos x + \cos(x + \frac{2\pi}{3}) + \cos(x - \frac{2\pi}{3}) = \mathrm{Re}\left(e^{ix}(1 + \omega + \omega^2)\right)$ with $\omega = e^{2\pi i/3}$.

Hint 2. $1+\omega+\omega^2=0$, so the integrand is identically 0 and the integral is a constant C.

Problem 7

Hint 1. Apply the king's rule to $I=\int_a^b \frac{f}{f+g}$: since $f(a+b-x)=g(x)$ and hence $g(a+b-x)=f(x)$, it becomes $\int_a^b \frac{g}{g+f}$. Add the two expressions.

Hint 2. $2I=b-a$. A classical example (Putnam 1987): $\int_2^4 \frac{\sqrt{\ln(9-x)}}{\sqrt{\ln(9-x)}+\sqrt{\ln(x+3)}}dx=1$, here $a+b=6$ and $f(x)=\sqrt{\ln(9-x)}$, $g(x)=\sqrt{\ln(x+3)}$; no elementary antiderivative is known (and none is expected).

Problem 8

Hint 1. Simplify each summand for non-integer x (integers form a null set): $\lfloor\lceil x\rceil\rfloor=\lceil x\rceil$, $\lceil\lfloor x\rfloor\rceil=\lfloor x\rfloor$, $\lfloor\{x\}\rfloor=0$, $\{\lfloor x\rfloor\}=0$, $\lceil\{x\}\rceil=1$, $\{\lceil x\rceil\}=0$.

Hint 2. The integrand is $\lfloor x\rfloor+\lceil x\rceil+1=2\lfloor x\rfloor+2$ almost everywhere, so the integral is $2\sum_{n=0}^{999} n+2000=1001000$.

Problem 9

Hint 1. $\int_0^N \binom{\lfloor x\rfloor}{k}dx=\sum_{n=0}^{N-1}\binom{n}{k}$, and the hockey-stick identity gives $\sum_{n=0}^{N-1}\binom{n}{k}=\binom{N}{k+1}$.

Hint 2. So the answers are $\binom{20}{3}=1140$ and $\binom{2026}{4}=699938073450$.

Problem 10

Hint 1. Either note that $F(\sin x)$, with $F'=f$, is an antiderivative and is 2π-periodic, or use $x\mapsto\pi-x$ (together with periodicity): sin is unchanged and cos changes sign.

Hint 2. General criterion: if a substitution $x\mapsto c-x$ maps the interval onto itself, or $x\mapsto x+T$ maps it onto itself modulo a period, and turns the integrand h into $-h$, then $I=-I$, so $I=0$. Good test cases from the class sheet: 2(b) with $x\mapsto\pi-x$, 2(c) with $x\mapsto 2\pi-x$, 2(d) with $x\mapsto\pi-x$, and the odd integrands in Problem 4.